

MATH GU4053 INTRO TO ALGEBRAIC TOPOLOGY

IN-CLASS MIDTERM, 2:40-3:55 ON 3/2/26

6 numbered problems

Closed-book; no notes, computers, calculators, nor visible phones

Unless explicitly stated otherwise, you may use any result proved in class. If your solution makes use of a theorem from class, make sure to explain why it applies (e.g., if the theorem has hypotheses).

Solutions based on an advanced result which has not been covered in this class, or one of its prerequisites, will not receive full credit. (Especially if the use of an advanced result significantly changes the difficulty of the problem.)

It is not sufficient to state that a problem is identical to one from a previous homework or practice exam, when that is the case.

If you run out of space, use the back side of the paper and indicate this clearly.

Name: _____

Uni: _____

1. (a) Let X and Y be topological spaces, and let $f_0, f_1 : X \rightarrow Y$ be two continuous maps. State the definition of what it means for f_0 and f_1 to be homotopic. Then state what it means for f_0 and f_1 to be homotopic relative to A , for a subset $A \subset X$.

- (b) Prove that being homotopic relative to A is an equivalence relation on the set of all continuous maps.
You should explain why the relation is transitive, but you may skip symmetry and reflexivity.

- (c) Let $y \in Y$. State the definition of $\pi_1(Y, y)$.
Give a correct definition of $\pi_1(Y, y)$ as a set. In this problem you do not need to explain the group structure.

- (d) State the definition of what it means for Y to be simply connected.

2. (a) State the definition of what it means for $p : X \rightarrow Y$ to be a covering map.

(b) True or false: if $S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$ and $p : S^1 \rightarrow [-1, 1]$ is given by $p((x, y)) = x$, then p is a covering map. *Justify your answer.*

(c) State the path lifting theorem for covering maps $p : X \rightarrow Y$.

3. (a) Let $p : X \rightarrow Y$ be a covering map, and let $F = p^{-1}(\{y\})$ for some $y \in Y$. Explain how the action of $\pi_1(Y, y)$ on F is defined.

- (b) Does there exist a covering map $f : K \rightarrow T$, where K is the Klein bottle and T is the torus?
Justify your answer.

4. Let $v_0 \in \mathbb{R}^2$, let $X = \mathbb{R}^2$ in the usual Euclidean topology, and define an action $\mathbb{Z} \times X \rightarrow X$ as $n.v = v + nv_0$.
- (a) Prove that the action is a covering space action if $v_0 \neq 0$, and determine the fundamental group of the orbit space $X/\mathbb{Z}$ in that case.

- (b) Prove that the action is not a covering space action when $v_0 = 0$.

5. (a) Let $f : (X, x) \rightarrow (Y, y)$ be a based map. State the definition of the induced homomorphism

$$f_* : \pi_1(X, x) \rightarrow \pi_1(Y, y).$$

Write a correct formula for f_ , stating clearly whether it depends on any choices. You do not need to prove that it is well defined.*

- (b) Let Z be a path connected topological space, and let $z_0, z_1 \in Z$. In class, we proved that the fundamental groups $\pi_1(Z, z_1)$ and $\pi_1(Z, z_0)$ are isomorphic. Explain how the isomorphism $\pi_1(Z, z_1) \rightarrow \pi_1(Z, z_0)$ is defined.

Write a correct formula for the isomorphism, stating clearly whether it depends on any choices. You do not need to prove that it is well defined, nor that it is an isomorphism.

6. For each of the following statements, mark if they are true or false.

In this problem, you do not need to justify your answers.

- (a) $\mathbb{R}^2 \setminus \{0\}$ is simply connected.
- (b) $\mathbb{R}^3 \setminus \{0\}$ is simply connected.
- (c) If X and Y are homeomorphic, then they are also homotopy equivalent.
- (d) If X and Y are homotopy equivalent, then they are also homeomorphic.
- (e) Any continuous $f : S^1 \rightarrow S^1$ is homotopic to the identity map.
- (f) Any covering map $p : S^1 \rightarrow S^1$ is homotopic to the identity map.
- (g) If U and V are simply connected open subsets of X , and $X = U \cup V$, then X is also simply connected.
- (h) If X is any topological space, and Y is contractible, then any continuous map $f : X \rightarrow Y$ is homotopic to a constant map.
- (i) Let $G \times X \rightarrow X$ be a covering space action and assume X is path connected, and that X/G is simply connected. Then $G = \{e\}$.
- (j) Let $G \times X \rightarrow X$ be a covering space action and assume that both X and X/G are simply connected. Then $G = \{e\}$.
- (k) There exists a covering map $p : S^1 \rightarrow S^2$.